

Movie_data_analysis

November 3, 2025

```
[60]: import pandas as pd
import numpy as np
import seaborn as sns

import matplotlib.pyplot as plt
import matplotlib.mlab as mlab
import matplotlib
plt.style.use('ggplot')
from matplotlib.pyplot import figure

%matplotlib inline
matplotlib.rcParams['figure.figsize'] = (12,8)

pd.options.mode.chained_assignment = None

# Now we need to read in the data
df = pd.read_csv(r'movies.csv')
```

```
[61]: # Now let's take a look at the data

df.head()
```

```
[61]:
```

	name	rating	genre	year	\
0	The Shining	R	Drama	1980	
1	The Blue Lagoon	R	Adventure	1980	
2	Star Wars: Episode V - The Empire Strikes Back	PG	Action	1980	
3	Airplane!	PG	Comedy	1980	
4	Caddyshack	R	Comedy	1980	

	released	score	votes	director	\
0	June 13, 1980 (United States)	8.4	927000.0	Stanley Kubrick	
1	July 2, 1980 (United States)	5.8	65000.0	Randal Kleiser	
2	June 20, 1980 (United States)	8.7	1200000.0	Irvin Kershner	
3	July 2, 1980 (United States)	7.7	221000.0	Jim Abrahams	
4	July 25, 1980 (United States)	7.3	108000.0	Harold Ramis	

	writer	star	country	budget \
0	Stephen King	Jack Nicholson	United Kingdom	19000000.0
1	Henry De Vere Stacpoole	Brooke Shields	United States	4500000.0
2	Leigh Brackett	Mark Hamill	United States	18000000.0
3	Jim Abrahams	Robert Hays	United States	3500000.0
4	Brian Doyle-Murray	Chevy Chase	United States	6000000.0

	gross	company	runtime
0	46998772.0	Warner Bros.	146.0
1	58853106.0	Columbia Pictures	104.0
2	538375067.0	Lucasfilm	124.0
3	83453539.0	Paramount Pictures	88.0
4	39846344.0	Orion Pictures	98.0

```
[62]: # Let's see if there is any missing data

for col in df.columns:
    pct_missing = np.mean(df[col].isnull())
    print('{} - {}'.format(col, round(pct_missing*100)))
```

```
name - 0%
rating - 1%
genre - 0%
year - 0%
released - 0%
score - 0%
votes - 0%
director - 0%
writer - 0%
star - 0%
country - 0%
budget - 28%
gross - 2%
company - 0%
runtime - 0%
```

```
[63]: # Data Types for our columns

print(df.dtypes)
```

```
name          object
rating         object
genre          object
year          int64
released       object
score         float64
votes         float64
```

```

director    object
writer      object
star        object
country     object
budget      float64
gross       float64
company     object
runtime     float64
dtype: object

```

```

[64]: # change data type for column
df = df.dropna()

df['budget'] = df['budget'].astype('int64')

df['gross'] = df['gross'].astype('int64')

```

```

[65]: # Create correct year column

df['yearcorrect'] = df['released'].astype(str).str.extract(r'(\d{4})')

```

```

[66]: # Order our Data a little bit to see

df = df.sort_values(by=['gross'], inplace=False, ascending=False)

```

```

[67]: # Drop duplicates and clean budget/gross columns

df.drop_duplicates()

```

```

[67]:

```

	name	rating	genre	year	\
5445	Avatar	PG-13	Action	2009	
7445	Avengers: Endgame	PG-13	Action	2019	
3045	Titanic	PG-13	Drama	1997	
6663	Star Wars: Episode VII - The Force Awakens	PG-13	Action	2015	
7244	Avengers: Infinity War	PG-13	Action	2018	
...	...	...	...	...	
5640	Tanner Hall	R	Drama	2009	
2434	Philadelphia Experiment II	PG-13	Action	1993	
3681	Ginger Snaps	Not Rated	Drama	2000	
272	Parasite	R	Horror	1982	
3203	Trojan War	PG-13	Comedy	1997	

	released	score	votes	\
5445	December 18, 2009 (United States)	7.8	1100000.0	
7445	April 26, 2019 (United States)	8.4	903000.0	
3045	December 19, 1997 (United States)	7.8	1100000.0	
6663	December 18, 2015 (United States)	7.8	876000.0	

7244	April 27, 2018 (United States)	8.4	897000.0
...	...	...	...
5640	January 15, 2015 (Sweden)	5.8	3500.0
2434	June 4, 1994 (South Korea)	4.5	1900.0
3681	May 11, 2001 (Canada)	6.8	43000.0
272	March 12, 1982 (United States)	3.9	2300.0
3203	October 1, 1997 (Brazil)	5.7	5800.0

	director	writer	star	\
5445	James Cameron	James Cameron	Sam Worthington	
7445	Anthony Russo	Christopher Markus	Robert Downey Jr.	
3045	James Cameron	James Cameron	Leonardo DiCaprio	
6663	J.J. Abrams	Lawrence Kasdan	Daisy Ridley	
7244	Anthony Russo	Christopher Markus	Robert Downey Jr.	
...	...	...	...	
5640	Francesca Gregorini	Tatiana von Fürstenberg	Rooney Mara	
2434	Stephen Cornwell	Wallace C. Bennett	Brad Johnson	
3681	John Fawcett	Karen Walton	Emily Perkins	
272	Charles Band	Alan J. Adler	Robert Glaudini	
3203	George Huang	Andy Burg	Will Friedle	

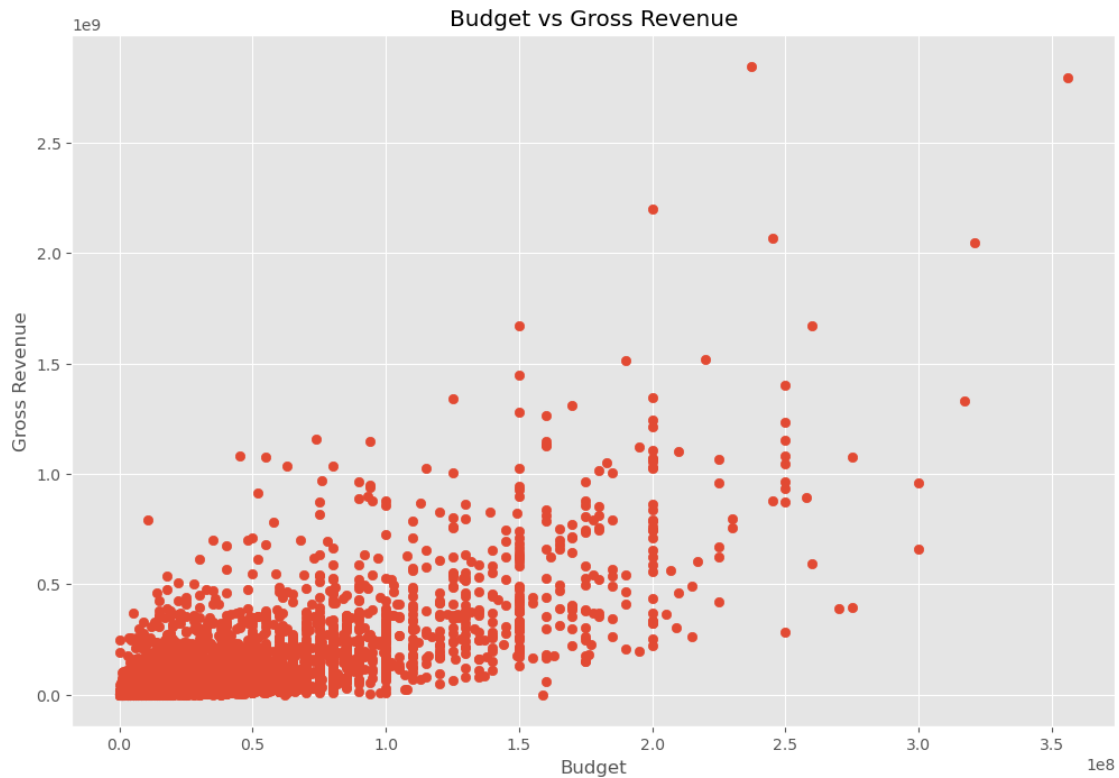
	country	budget	gross	company	\
5445	United States	237000000	2847246203	Twentieth Century Fox	
7445	United States	356000000	2797501328	Marvel Studios	
3045	United States	200000000	2201647264	Twentieth Century Fox	
6663	United States	245000000	2069521700	Lucasfilm	
7244	United States	321000000	2048359754	Marvel Studios	
...	...	...	...	...	
5640	United States	3000000	5073	Two Prong Lesson	
2434	United States	5000000	2970	Trimark Pictures	
3681	Canada	5000000	2554	Copperheart Entertainment	
272	United States	800000	2270	Embassy Pictures	
3203	United States	15000000	309	Daybreak	

	runtime	yearcorrect
5445	162.0	2009
7445	181.0	2019
3045	194.0	1997
6663	138.0	2015
7244	149.0	2018
...	...	...
5640	96.0	2015
2434	97.0	1994
3681	108.0	2001
272	85.0	1982
3203	85.0	1997

[5421 rows x 16 columns]

```
[68]: # Scatter plots with budget vs gross

plt.scatter(x=df['budget'], y=df['gross'])
plt.title('Budget vs Gross Revenue')
plt.xlabel('Budget')
plt.ylabel('Gross Revenue')
plt.show()
```



```
[69]: # Plot budget vs gross using seaborn
print(df.dtypes)
sns.regplot(x='budget', y='gross', data=df_clean, scatter_kws={"color": "red"},
↪ line_kws={"color": "blue"})
```

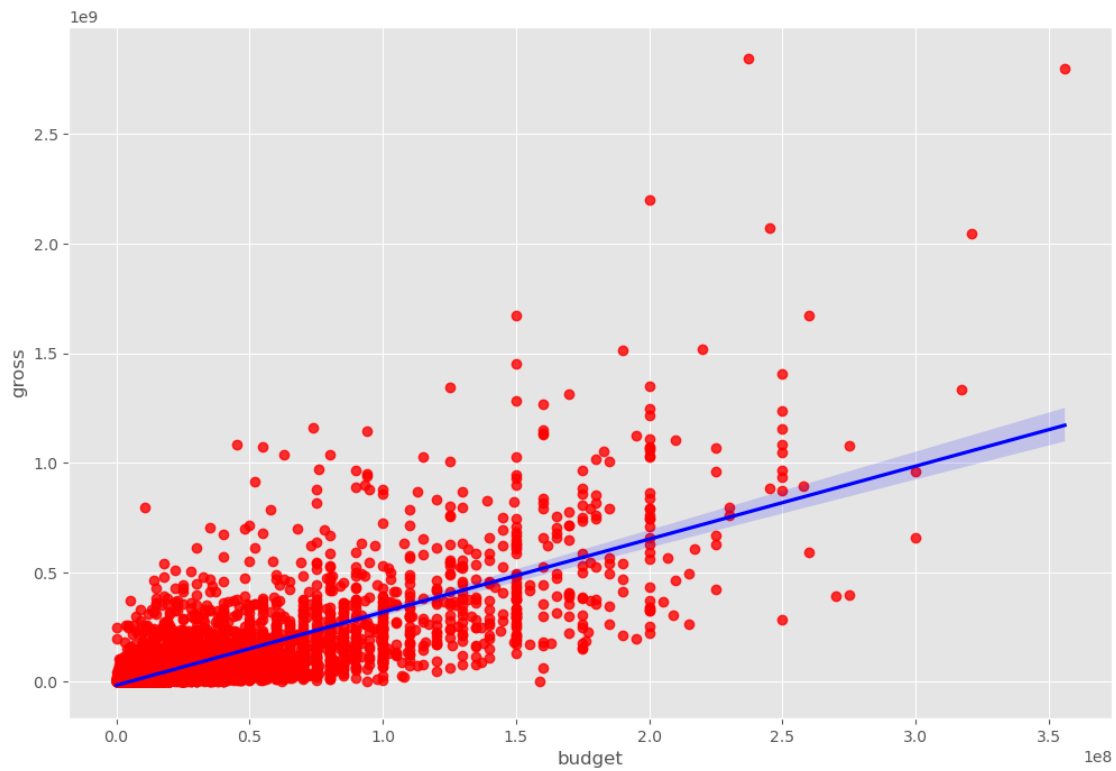
name	object
rating	object
genre	object
year	int64
released	object
score	float64
votes	float64

```

director      object
writer        object
star          object
country       object
budget        int64
gross         int64
company       object
runtime       float64
yearcorrect   object
dtype: object

```

```
[69]: <Axes: xlabel='budget', ylabel='gross'>
```



```
[73]: # Correlations
numeric_df = df.select_dtypes(include=['float64', 'int64', 'Int64'])
corr_matrix = numeric_df.corr()
print(corr_matrix)
```

	year	score	votes	budget	gross	runtime
year	1.000000	0.056386	0.206021	0.327722	0.274321	0.075077
score	0.056386	1.000000	0.474256	0.072001	0.222556	0.414068
votes	0.206021	0.474256	1.000000	0.439675	0.614751	0.352303
budget	0.327722	0.072001	0.439675	1.000000	0.740247	0.318695

```
gross    0.274321  0.222556  0.614751  0.740247  1.000000  0.275796
runtime  0.075077  0.414068  0.352303  0.318695  0.275796  1.000000
```

```
[75]: # High correlation between gross and budget

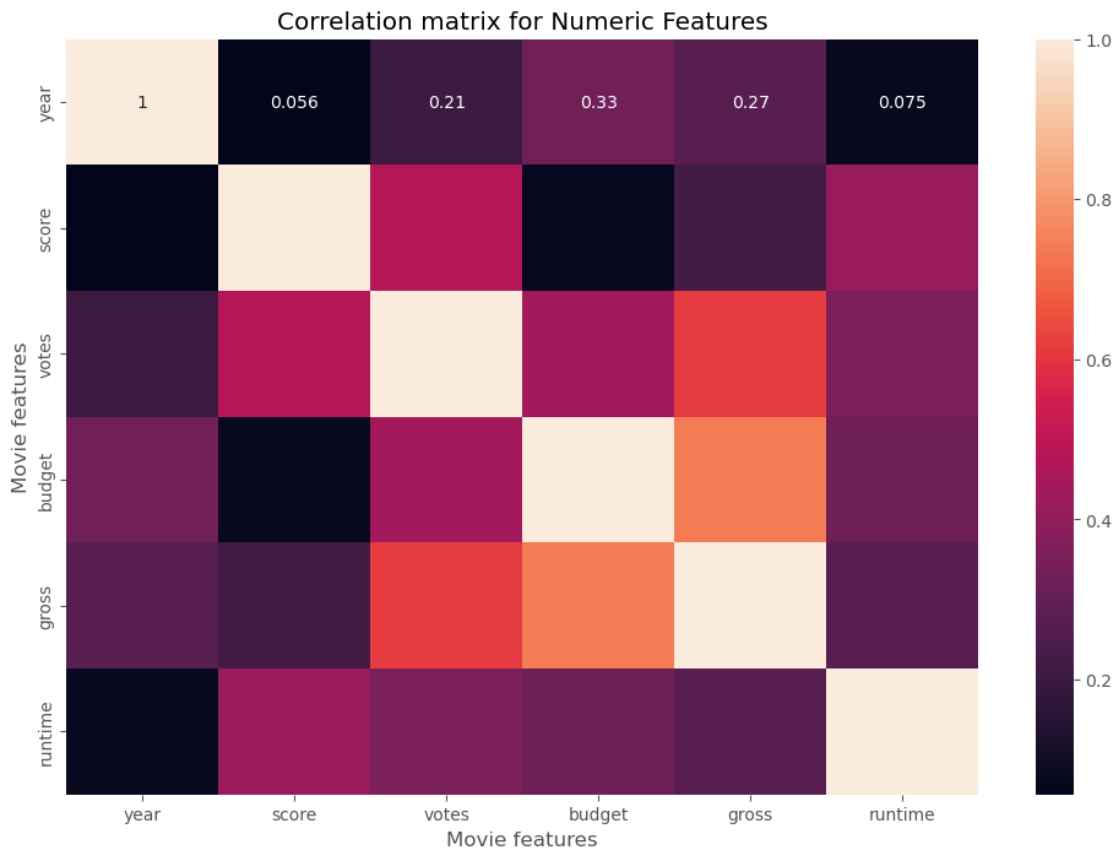
sns.heatmap(corr_matrix, annot = True)

plt.title("Correlation matrix for Numeric Features")

plt.xlabel("Movie features")

plt.ylabel("Movie features")

plt.show()
```



```
[76]: df_numerized = df

for col in df_numerized.columns:
    if df_numerized[col].dtype == 'object':
        df_numerized[col] = df_numerized[col].astype('category')
```

```
df_numerized[col] = df_numerized[col].astype('category').cat.codes
```

```
df_numerized
```

```
[76]:
```

	name	rating	genre	year	released	score	votes	director	writer	\
5445	386	5	0	2009	527	7.8	1100000.0	785	1263	
7445	388	5	0	2019	137	8.4	903000.0	105	513	
3045	4909	5	6	1997	534	7.8	1100000.0	785	1263	
6663	3643	5	0	2015	529	7.8	876000.0	768	1806	
7244	389	5	0	2018	145	8.4	897000.0	105	513	
...	...	...	...	...	...	...	...	...	...	...
5640	3794	6	6	2009	890	5.8	3500.0	585	2924	
2434	2969	5	0	1993	1467	4.5	1900.0	1805	3102	
3681	1595	3	6	2000	1721	6.8	43000.0	952	1683	
272	2909	6	9	1982	1525	3.9	2300.0	261	55	
3203	4966	5	4	1997	2152	5.7	5800.0	651	161	

	star	country	budget	gross	company	runtime	yearcorrect
5445	1534	47	237000000	2847246203	1382	162.0	29
7445	1470	47	356000000	2797501328	983	181.0	39
3045	1073	47	200000000	2201647264	1382	194.0	17
6663	356	47	245000000	2069521700	945	138.0	35
7244	1470	47	321000000	2048359754	983	149.0	38
...	...	...	...	...	...	...	...
5640	1498	47	3000000	5073	1385	96.0	35
2434	186	47	5000000	2970	1376	97.0	14
3681	527	6	5000000	2554	466	108.0	21
272	1473	47	800000	2270	582	85.0	2
3203	1811	47	15000000	309	504	85.0	17

```
[5421 rows x 16 columns]
```

```
[82]: # Correlations
print(df_numerized.corr())

sns.heatmap(df_numerized.corr(), annot = True)

plt.title("Correlation matrix")

plt.xlabel("Movie features")

plt.ylabel("Movie features")

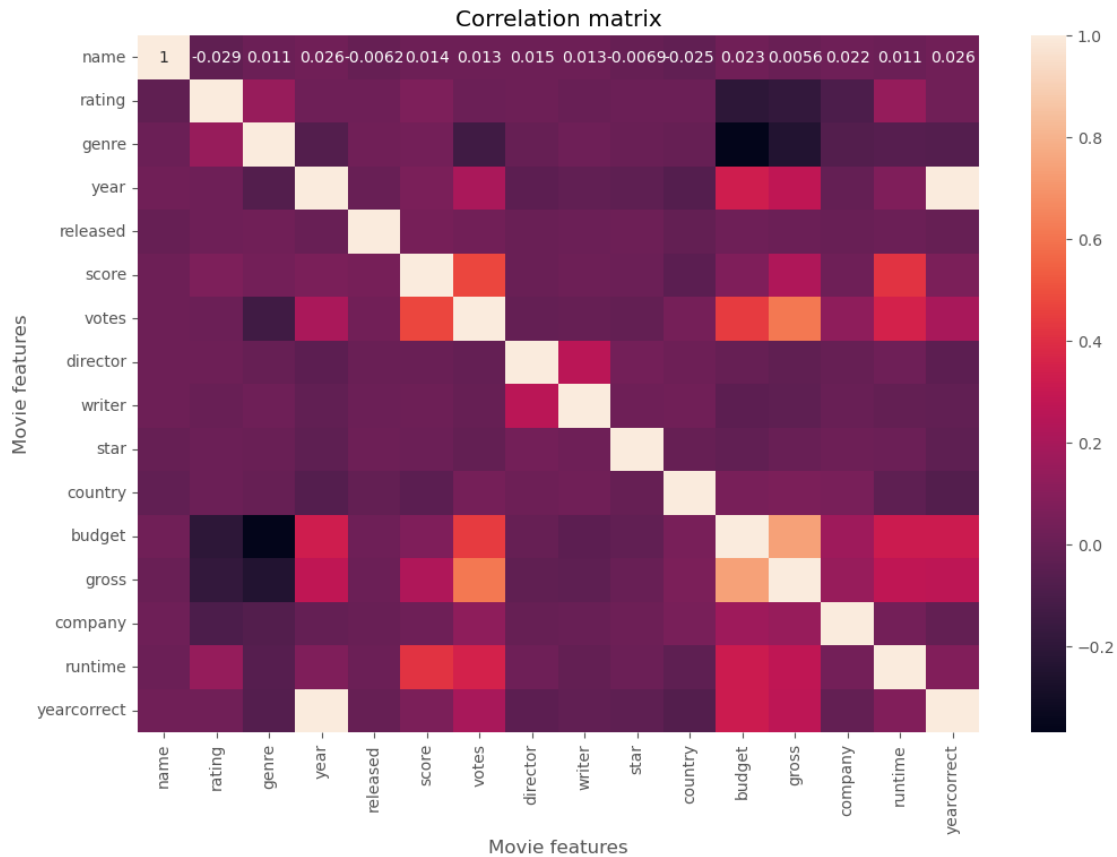
plt.show()
```

	name	rating	genre	year	released	score	votes	director	writer	\
5445	386	5	0	2009	527	7.8	1100000.0	785	1263	

7445	388	5	0	2019	137	8.4	903000.0	105	513
3045	4909	5	6	1997	534	7.8	1100000.0	785	1263
6663	3643	5	0	2015	529	7.8	876000.0	768	1806
7244	389	5	0	2018	145	8.4	897000.0	105	513
...	...	...	...	...	...	...	...	...	...
5640	3794	6	6	2009	890	5.8	3500.0	585	2924
2434	2969	5	0	1993	1467	4.5	1900.0	1805	3102
3681	1595	3	6	2000	1721	6.8	43000.0	952	1683
272	2909	6	9	1982	1525	3.9	2300.0	261	55
3203	4966	5	4	1997	2152	5.7	5800.0	651	161

	star	country		budget		gross	company	runtime	yearcorrect
5445	1534	47	237000000	2847246203	1382	162.0			29
7445	1470	47	356000000	2797501328	983	181.0			39
3045	1073	47	200000000	2201647264	1382	194.0			17
6663	356	47	245000000	2069521700	945	138.0			35
7244	1470	47	321000000	2048359754	983	149.0			38
...	...	...	...	...	...	...	...	...	...
5640	1498	47	3000000	5073	1385	96.0			35
2434	186	47	5000000	2970	1376	97.0			14
3681	527	6	5000000	2554	466	108.0			21
272	1473	47	800000	2270	582	85.0			2
3203	1811	47	15000000	309	504	85.0			17

[5421 rows x 16 columns]



```
[86]: mat_corr = df_numerized.corr()

corr_pairs = mat_corr.unstack()

print(corr_pairs)
```

```
[86]: name          name          1.000000
      rating        -0.029234
      genre         0.010996
      year          0.025542
      released      -0.006152
      ...
yearcorrect budget      0.320312
yearcorrect gross       0.268721
yearcorrect company     -0.018806
yearcorrect runtime      0.075294
yearcorrect yearcorrect  1.000000
Length: 256, dtype: float64
```

```
[87]: sorted_pairs = corr_pairs.sort_values()
      print(sorted_pairs)
```

```
budget    genre    -0.368523
genre     budget    -0.368523
          gross     -0.244101
gross     genre     -0.244101
budget    rating    -0.203946
...
votes     votes     1.000000
country   country   1.000000
rating    rating    1.000000
star      star      1.000000
director  director  1.000000
Length: 256, dtype: float64
```

```
[88]: # high correlation

high_corr = sorted_pairs[(sorted_pairs > 0.5)]
print(high_corr)
```

```
gross      votes      0.614751
votes      gross      0.614751
gross      budget     0.740247
budget     gross      0.740247
year       yearcorrect 0.998726
yearcorrect year      0.998726
writer     writer      1.000000
year       year      1.000000
released   released    1.000000
score      score      1.000000
budget     budget      1.000000
company    company     1.000000
genre      genre      1.000000
name       name        1.000000
yearcorrect yearcorrect 1.000000
gross      gross      1.000000
runtime    runtime     1.000000
votes      votes      1.000000
country    country     1.000000
rating     rating      1.000000
star       star        1.000000
director   director    1.000000
dtype: float64
```

```
[ ]: # votes and budgets have the highest correlation
```